

Section 3: Monotone Functional

1 Monotone Functional

1.1 Definition of the Functional

We define the global energy functional $Q(t)$ as follows:

$$Q(t) = \|u(t)\|_{L^2}^2 + \|\nabla u(t)\|_{L^2}^2,$$

where:

- $\|u(t)\|_{L^2}^2 = \int_{\mathbb{R}^3} |u(x, t)|^2 dx$ is the kinetic energy of the flow.
- $\|\nabla u(t)\|_{L^2}^2 = \int_{\mathbb{R}^3} |\nabla u(x, t)|^2 dx$ measures the enstrophy of the flow.

The functional $Q(t)$ combines the global energy and the gradient energy, capturing key quantities that influence the regularity of solutions to the Navier–Stokes equations.

1.2 Key Theorem: Boundedness of $Q(t)$

Theorem 1.1. *Under the assumptions of the Navier–Stokes equations and the Scale-Barrier Principle, the global energy functional $Q(t)$ remains bounded for all time $t > 0$:*

$$Q(t) \leq Q(0)e^{-\nu t} + \int_0^t (C\|\nabla u(s)\|_{L^2}^3 + 2\|f(s)\|_{L^2}\|u(s)\|_{L^2}) ds.$$

Proof. The boundedness of $Q(t)$ follows from the evolution equation:

$$\frac{dQ}{dt} = -\nu\|\nabla u\|_{L^2}^2 - \nu\|\Delta u\|_{L^2}^2 + C\|\nabla u\|_{L^2}^3 + 2 \int_{\mathbb{R}^3} f \cdot u dx.$$

Using the Scale-Barrier Principle, which ensures rapid spectral decay at high wave numbers, we control the nonlinear term $C\|\nabla u\|_{L^2}^3$ and dissipation term $\nu\|\Delta u\|_{L^2}^2$. The forcing term $\int f \cdot u$ is bounded using:

$$2 \int_{\mathbb{R}^3} f \cdot u dx \leq 2\|f\|_{L^2}\|u\|_{L^2}.$$

Substituting these bounds and integrating over time yields the stated result. $\square$

1.3 Energy Evolution

To derive the time evolution of $Q(t)$, we compute the derivatives of each term:

$$\begin{aligned}\frac{d}{dt}\|u(t)\|_{L^2}^2 &= 2 \int_{\mathbb{R}^3} u \cdot \frac{\partial u}{\partial t} dx, \\ \frac{d}{dt}\|\nabla u(t)\|_{L^2}^2 &= 2 \int_{\mathbb{R}^3} \nabla u : \nabla \frac{\partial u}{\partial t} dx.\end{aligned}$$

Substituting the Navier–Stokes equations:

$$\frac{\partial u}{\partial t} + (u \cdot \nabla)u + \nabla p - \nu \Delta u = f,$$

and integrating by parts, we obtain:

$$\begin{aligned}\frac{d}{dt}\|u(t)\|_{L^2}^2 &= -2\nu\|\nabla u\|_{L^2}^2 + 2 \int_{\mathbb{R}^3} f \cdot u dx, \\ \frac{d}{dt}\|\nabla u(t)\|_{L^2}^2 &= -2\nu\|\Delta u\|_{L^2}^2 + C\|\nabla u\|_{L^2}^3,\end{aligned}$$

where $C\|\nabla u\|_{L^2}^3$ arises from the nonlinear term $(u \cdot \nabla)u$.

Combining these results, the time evolution of $Q(t)$ becomes:

$$\frac{dQ}{dt} = -\nu\|\nabla u\|_{L^2}^2 - \nu\|\Delta u\|_{L^2}^2 + C\|\nabla u\|_{L^2}^3 + 2 \int_{\mathbb{R}^3} f \cdot u dx.$$

1.4 Integration with the Scale-Barrier Principle

The Scale-Barrier Principle ensures that the energy spectrum $E(k, t)$ decays rapidly at high wave numbers k , preventing the unbounded growth of velocity gradients. Specifically:

- The dissipation term $-\nu\|\Delta u\|_{L^2}^2$ dominates at high wave numbers, ensuring energy cannot cascade indefinitely.
- The nonlinear term $C\|\nabla u\|_{L^2}^3$ is controlled due to the boundedness of $\|\nabla u\|_{L^2}$ ensured by the saturation effect of the Scale-Barrier Principle.

1.5 Extensions to Fractional Sobolev Norms

To enhance the applicability of $Q(t)$, we propose extending it to fractional Sobolev spaces H^s for $s \in \mathbb{R}$:

$$Q_s(t) = \|u(t)\|_{L^2}^2 + \|(-\Delta)^{s/2}u(t)\|_{L^2}^2,$$

where $\|(-\Delta)^{s/2}u\|_{L^2}$ captures intermediate and high-frequency contributions. This generalization:

- Provides finer control over the spectral energy distribution.
- Captures nonlocal effects crucial in turbulence modeling.

Challenges:

- Deriving fractional evolution equations that extend the standard Navier–Stokes formulation.
- Ensuring compactness in fractional norms, which requires stronger assumptions on initial data.

1.6 Adaptation to Irregular Domains

Real-world fluid flows often occur in irregular or bounded domains. To extend the functional $Q(t)$ to such settings:

- Define the energy functional on bounded domains Ω with appropriate boundary conditions:

$$Q_\Omega(t) = \|u(t)\|_{L^2(\Omega)}^2 + \|\nabla u(t)\|_{L^2(\Omega)}^2.$$

- Incorporate boundary terms to account for energy flux and dissipation near the boundaries.

Key Challenges:

- Compatibility of boundary conditions (e.g., Dirichlet or Navier slip) with dissipation mechanisms.
- Addressing energy accumulation near irregular boundaries.

1.7 Numerical Validation Framework

To validate the boundedness of $Q(t)$ and its extensions numerically:

1. **Setup**: Solve the Navier–Stokes equations using Fourier spectral methods for periodic domains or finite-element methods for irregular domains.
2. **Goals**:
 - Track the evolution of $Q(t)$, $Q_s(t)$, and $Q_\Omega(t)$.
 - Confirm dissipation dominance over nonlinear growth.
3. **Visualization**: Generate plots for:
 - Time evolution of $Q(t)$ showing bounded behavior.
 - Energy contributions from dissipation and nonlinear terms.

1.8 Integration with Compactness and Regularity

The boundedness of $Q(t)$ plays a critical role in ensuring compactness of approximate solutions:

- The bounded velocity gradient $\|\nabla u\|_{L^2}$ ensures convergence in H^1 norms.
- Strong convergence in L^2 is achieved due to dissipation control on $\|\Delta u\|_{L^2}$.

1.9 Historical Context and Broader Implications

Comparison with Previous Functionals: The functional $Q(t)$ refines traditional energy approaches by incorporating enstrophy bounds and enabling extensions to fractional norms and irregular domains.

Future Directions: Extending to fractional Sobolev spaces or irregular domains enhances the framework's applicability to real-world and theoretical scenarios.

References

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